RED FM

Weekly checks

Border Holdings, Avonmouth

Date: 2026-10-28

Ref:	WK-MVS2ATD2
Date:	2026-10-28
Engineer:	Dexter Norley
Plant:	A & B, C & D, E
Week:	44
Ambient °C:	13.8
Faults recorded:	0

Faults recorded on this visit

None - no line item was flagged.

A & B - Chamber temperature

Item	A1	A2	A3	A4	No1 B2	No2 B2	No3 B1	No4 B1
Cold store set point (controller)	-20	-20	-20	-20	-18.9	-18.9	-18.9	-18.9
Cold store temp air-on	-19.9	-19.8	-20.1	-20.2	-17.2	-17.4	-17.8	-18.1
Coolers air off	-21.9	-21.7	-21.5	-21.1	-18.9	-18.8	-19	-19.5
Cooling on/off	On	On	On	On	On	On	On	On

A & B - Coolers

Item	A1	A2	A3	A4	No1 B2	No2 B2	No3 B1	No4 B1
Cooler fan correct operation	OK	OK	OK	OK	OK	OK	OK	OK
Cooler correct operation	OK	OK	OK	OK	OK	OK	OK	OK
Cooler correct defrosting cycle	OK	OK	OK	OK	OK	OK	OK	OK
Cooler ice build-up & debris	OK	OK	OK	OK	OK	OK	OK	OK
Defrost drain line for leakage	OK	OK	OK	OK	OK	OK	OK	OK
Operating suction superheat #	8	8.2	9.2	7.7	12	6.4	8.5	9.5
% opening of expansion valve #	48	56	29	26	100	5	23	100
Cooler air temp diff #	2c	1.9	1.4	0.9	1.7	1.4	1.2	1.4

A & B - Compressor racks & condenser

Item	A	B
Compr abnormal noise / vibration	OK	OK
Oil cooler fin face free of debris	OK	OK
Oil cooler fans correct operation	OK	OK
Plant-room ventilation fan (on/off)	Off	Off
Plant-room temperature	22.4	22.4
Wet-matt cooling on/off (hours)	Off	Off
System delivery pressure	12.8	12.1bar
Condenser fin face free of debris	OK	OK
Condenser fans correct operation	Fan 3 u/s	Yes
Liquid line sight glass	Clear	Clear
Receiver level	Above bottom sight glass	Above bottom sight glass
Oil separator oil level	Full	Full
No1 compressor running & % load	0%	0%
No2 compressor running & % load	100%	75%

Item	A	B
No3 compressor running & % load	0%	-
No1 compr delivery & econ temp	Off	Off
No2 compr delivery & econ temp	Delivery 66 econ 5c	Delivery 58.4c econ off
No3 compr delivery & econ temp	Off	-
Refrigerant leak detection system	OK	OK

A & B - Controls

Item	A	B
Suction pressure	0.8bar	1bar
Suction temperature	-14.2c	-14c
Suction SH	13c	10c
Delivery pressure	12bar	11.8bar
Common delivery temp	66c	58c
Oil temperature	47.8c	35c
HP liquid temperature	24c	23c
HP liquid sub-cooling	5c	6c
Econ liquid temperature	4c	21c
Econ sub-cooling	20c	8c

A & B - Main incoming meter

Item	kWhr
Electric kWhr meter reading	716614
Previous meter reading	712195
Total kWhr	4419

A & B - A - pack meter

Item	kWhr
Electric kWhr meter reading	4686195
Previous meter reading	4676174
Pack total kWhr	10021
A chamber total energy	13624

A & B - B - pack meter

Item	kWhr
Electric kWhr meter reading	2904816
Previous meter reading	2897251
Pack total kWhr	7565
B chamber total energy	10467

A & B - A & B cooler energy meter

Item	kWhr
Electric kWhr meter reading	2671909
Previous meter reading	2664804
Total kWhr	7105

A & B - B cooler energy meter

Item	kWhr
Electric kWhr meter reading	724917
Previous meter reading	721415
Total kWhr	2902

A & B - A cooler energy (derived)

Item	kWhr
Electric kWhr meter reading	1946992
Previous meter reading	1943389
Total kWhr	3603

C & D - Chamber temperature

Item	C1	C2	C3	C4	D1	D2
Cold store set point (controller)	-18.8	-18.8	-18.8	-18.8	-18.6	-18.1
Cold store temp air-on	-18.2	-18.8	-18.9	-18.1	-18.8	-18.5
Coolers air off	-21.7	-20.7	-20.3	-19.6	-21.2	-21
Cooling on/off	On	On	On	On	On	On

C & D - Condensing units

Item	C1	C2	C3	C4	D1	D2	D3	D4
Compr abnormal noise / vibration	OK	OK	OK	OK	OK	OK	OK	OK
Compressor abnormal frosting	OK	OK	OK	OK	OK	OK	OK	OK
Compr oil level	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3
Compr delivery temperature	82	80	79	80	79	78	77	77
Oil recovery / separator level	Full	Full	Full	Full	Full	Full	Full	Full
Condenser fin face free of debris	OK	OK	OK	OK	OK	OK	OK	OK
Condenser fans abnormal noise	OK	OK	OK	OK	OK	OK	OK	OK
Liquid line sight glass	OK	OK	OK	OK	OK	OK	OK	OK
Receiver level	Below bottom sight glass	Above bottom sight glass	Below bottomsight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass

C & D - Coolers

Item	C1	C2	C3	C4	D1	D2
Cooler temperature diff	3.5	1.9	1.4	1.5	2.4	2.5
Cooler correct operation	OK	OK	OK	OK	OK	OK
Cooler correct defrosting cycle	OK	OK	OK	OK	OK	OK
Cooler ice build-up & debris	OK	OK	OK	OK	OK	OK
Defrost drain line for leakage	OK	OK	OK	OK	OK	OK

C & D - Energy readings

Item	C2-4 refrig	C1-3 refrig	D 1&2	D 3&4 compr
Electric kWhr meter reading	3040771	1140720	2670599	24653
Previous kWhr meter reading	3035777	1138753	2664530	24614
Total kWhr	4994	1967	6069	39
Chamber total energy		6961		6108

E - Chamber temperature

Item	E1	E2
Cold store set point (controller)	-18.6	-18.6
Cold store temp air-on	-18.6	-18.5
Coolers air off	-19.8	-19.7
Cooling on/off	On	On

E - E chamber front-centre temp

Item	°C
Front-centre temperature	-19.3

E - Condensing units

Item	E1 Compr1	E1 Compr2	E2 Compr1	E2 Compr2
Compr abnormal noise / vibration	OK	OK	OK	OK
Compressor abnormal frosting	OK	OK	OK	OK
Compr oil level	2/3	2/3	2/3	2/3
Compr delivery temperature	68	69	69	69
Oil recovery / separator level	Full	Full	Full	Full
Condenser fin face free of debris	OK	OK	OK	OK
Condenser fans abnormal noise	OK	OK	OK	OK
Liquid line sight glass	OK	OK	OK	OK

E - Coolers

Item	E1	E2
Cooler temperature diff	1.2	1.2
Cooler correct operation	OK	OK
Cooler correct defrosting cycle	OK	OK
Cooler ice build-up & debris	OK	OK
Defrost drain line for leakage	OK	OK

E - System readings

Item	E1	E2
Suction pressure	0.8	0.8
Suction temperature	-17	-16
Suction superheat	10	11
Delivery pressure	13	13.8
Liquid refrig temperature	26	27
Liquid sub	7	8

E - Energy readings

Item	E1	E2
Electric kWhr meter reading	14487	12786
Previous kWhr meter reading	14431	12760
Total kWhr	56	26
Chamber total energy		82

RDM alarms at time of visit

Item	Asset / chamber	Active since	Outcome	Action taken / reason
So2 stage fault	A pack	2026-10-28	Ongoing	New condensers installed in January